

T. Sai Rohan Reddy

✉ smecrohanreddy5p6@gmail.com | 📞 +91 76718 56610 | [🌐 LinkedIn](#) | [🐙 GitHub](#) | [🌐 Portfolio](#)

SUMMARY

Full-Stack Developer with hands-on experience building secure, scalable web applications using the MERN stack. Skilled in RESTful APIs, JWT authentication, RBAC, MongoDB, and SQL databases.

EDUCATION

St. Martin's Engineering College <i>B.Tech in Computer Science CGPA: 7.43</i>	Telangana <i>Expected 2026</i>
SR Junior College <i>Intermediate (Class XII) CGPA: 7.78</i>	Nizamabad <i>2022</i>
Victory High School <i>High School (Class X) CGPA: 8.5</i>	Chandhur <i>2020</i>

PROJECTS

PulpFibrilNet — CNN & Image Processing System for Pulp Fibril Detection

Tech Stack: Python, TensorFlow, ResNet-50, YOLOv4, OpenCV

[🐙 GitHub Repository](#)

- Designed a hybrid deep learning pipeline combining a fine-tuned ResNet-50 CNN with adaptive image processing (Bradley's local thresholding, morphological operations) to detect and quantify fibrils in microscopic pulp fiber images, replacing manual inspection.
- Integrated a YOLOv4-based post-processing stage to suppress false positives, achieving an average deviation of 0.449% against manual measurements across varying lighting, noise, and contrast conditions.
- Built an end-to-end analysis workflow covering preprocessing, model training/evaluation, and prediction visualization, delivering a scalable automated alternative to industrial fibrillation-index measurement.

WisperHub — Anonymous Social Media Platform

Tech Stack: Node.js, Express.js, MongoDB, JWT, Swagger UI, Rate Limiting

[🐙 GitHub Repository](#)

- Built a scalable anonymous social media REST API with posts, likes, comments, and follows using JWT cookie-based authentication.
- Engineered MongoDB multi-document transactions ensuring 100% data consistency; IP-based ban system with cron cleanup cut moderation effort by 70%.
- Enforced rate limiting (100 req/hr/client), reducing brute-force attacks by 40%; Swagger UI docs reduced onboarding time by 50%.

Skyo — Flight Booking Backend Architecture

Tech Stack: Node.js, Express.js, MongoDB, JWT, MVC, Repository Pattern, Nodemailer

[🐙 GitHub Repository](#)

- Built a flight booking backend with real-time seat availability tracking, eliminating overbooking and reducing search API response time by 35%.
- Applied MVC and Repository patterns with centralized error handling; OTP-based email verification via Nodemailer increased registration completion by 50%.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java
Backend: Node.js, Express.js, REST API
Frontend: React, HTML5, CSS3, Tailwind CSS
Databases: MongoDB, MySQL, Redis
Data Science / ML: NumPy, Pandas, Matplotlib
Tools: Git, GitHub, Postman, Swagger UI, VS Code, Linux, Render

CERTIFICATIONS

Advanced Backend Development with Node.js — Coding Ninjas
Frontend Web Development — Coding Ninjas
Java with Data Structures & Algorithms — Coding Ninjas

ACHIEVEMENTS

6-Hour AI Hackathon, CMR College: Built AI-Guru, an AI-powered diet and exercise advisor using Generative AI and automation workflows.
24-Hour Hackathon, Mallareddy College: Built AgriGo, a MERN-based crop guidance platform integrated with Google Gemini API.